

CSPG4 tissue RNA enrichment in extraskeletal myxoid chondrosarcoma depends on comparator and sequencing year

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Running title. Tissue RNA priorities in EMC

Abstract

Background: Comparative tissue RNA analysis can prioritize molecular characterization of rare tumors, but results depend on the comparator and specimen context. We evaluated a fixed 11-gene panel in extraskeletal myxoid chondrosarcoma (EMC), with CHRNA6 as a separate RNA-marker control.

Methods: Public RNA-sequencing data provided nine primary EMC after excluding three previously reported cases and one recurrence. Comparators were myxoid liposarcoma (n=14), low-grade fibromyxoid sarcoma (LGFMS; n=13) and synovial sarcoma (n=18). We calculated probability of superiority, $A=P(\text{EMC}>\text{comparator})+0.5P(\text{tie})$, with equal histology weights, both marginally and within sequencing years. Each matched contrast contained three EMC; their union comprised four patients. The prioritization rule was frozen before new target values were inspected. Original GSE24369 arrays provided a separate LGFMS comparison (6 EMC biopsies; 17 LGFMS).

Results: CSPG4 alone met the panel rule: marginal $A=0.895$ and matched $A=0.811$. Its LGFMS-specific A was 1.000 in arrays and 0.966 marginal/0.933 matched in RNA sequencing. Single-patient and single-histology deletions retained positive composite directions. Removing 2019, however, reversed the matched composite to 0.433. The marginal dermatofibrosarcoma protuberans (DFSP) context comparison was also nonpositive ($A=0.467$). CHRNA6 had $A=1$ in all primary comparisons. Other candidates showed weak, reversed or comparator-dependent effects.

Conclusions: CSPG4 warrants assessment of malignant-cell protein localization in EMC. Small matched samples, year dependence and normal-expression overlap limit generalization; these RNA results establish neither accessible cell-surface protein, diagnostic validation nor therapeutic benefit.

Keywords

Keywords. extraskeletal myxoid chondrosarcoma; CSPG4; CHRNA6; tissue RNA; comparator sensitivity; reproducibility

Introduction

Comparative expression data can identify a molecular feature worth examining in rare tumor tissue. The comparison matters: higher RNA expression against one sarcoma histology may disappear against another, and a bulk signal does not identify the cells that produced it. For extraskeletal myxoid chondrosarcoma (EMC), a useful analysis should therefore retain individual comparator histologies and technical sensitivities alongside the overall result.

EMC illustrates this distinction. Dulken and colleagues identified CHRNA6 through expression-array analysis of GSE24369 and subsequently demonstrated strong, diffuse CHRNA6 **RNA chromogenic in situ hybridization (CISH)** in 25 EMC cases. Their mimic series contained no threshold-level overexpression, although limited below-threshold signal occurred in 69 of 685 mimics. This was an RNA tissue assay, not protein immunohistochemistry or a therapeutic-accessibility experiment.[1] CHRNA6 is consequently a prior-supported context control here, not a newly discovered address; reanalysis of its discovery array is not an independent validation of that discovery.

We examined whether 11 previously nominated transcripts show consistent EMC-high ordering across specified sarcoma histologies in a public tissue RNA-sequencing cohort. We then compared the same histology across that cohort and original public arrays. This design makes the sensitivity to comparator choice explicit and asks a concrete follow-up question: does the RNA evidence justify examining CSPG4 protein localization in EMC tissue? Normal-expression observations provide separate context for interpreting that question.

Methods

Cohorts and fixed panel

Hofvander et al. released a 19,116-gene by 704-patient TPM matrix with sample metadata and supplementary clinicopathological information.[2] The study primarily recruited patients diagnosed and treated in Lund or Stockholm during 1988–2020, with some specimens from earlier collaborations. Gene quantification used STAR 2.7.10b/GRCh38 and RSEM 1.3.3/Ensembl 104. We retained original Table S1 diagnoses; transcriptomic reclassification was examined separately. Of 13 EMC specimens, three explicitly previously reported cases were excluded using the source's case references (104-92, 168-97, 536-00), as was one local recurrence (5081-14). Nine primary specimens remained. All nonblank recurrence/metastasis specimen flags were excluded symmetrically in comparator histologies. Positive overlap evidence was removed; absence of a publication reference was not interpreted as proof of universal independence. These exclusions

support an overlap-reduced comparison, with probable rather than completely verified independence from all historical datasets.

The panel was carried forward from the research program's curated list of transcripts named by proposed cell-directed, stromal and peptide-HLA approaches. Its documented source includes coverage corrections for stromal targets and CSPG4; it is not an exhaustive surfaceome [3]. The exact list and decision rule were frozen before the new cohort's target values were inspected [4]. This was prospective to the investigator's analysis of those values, not public preregistration or prospective specimen collection. The prespecified address panel was CD276, SSTR2, PRAME, FAP, CD248, CSPG4, MSLN, L1CAM, GPC3, ALPP and CDH17. CHRNA6 was analyzed separately. PRAME represents an intracellular peptide-HLA address concept rather than an ordinary intact surface protein; no endogenous peptide presentation was measured. Each symbol had one exact gene row in the TPM matrix. No expression-dependent sample or gene filtering was applied.

The primary same-cohort comparators were myxoid liposarcoma (MLPS), LGFMS and synovial sarcoma, with equal weights. Myxofibrosarcoma (MFS) and dermatofibrosarcoma protuberans (DFSP) were separate context comparisons. The original GSE24369 GPL6244 dataset comprised 42 samples: 6 EMC biopsies, 17 LGFMS, 6 MFS, 6 desmoids, 5 solitary fibrous tumors (SFT) and 2 pooled skeletal-muscle RNAs.[5] All original biopsies were retained because lesion stage was not established; they were not assumed to be primary lesions. The two normal pools were descriptive context, not two individual healthy controls. One uniquely assigned transcript cluster per gene was selected from original platform annotation before inspecting target values. Released RMA log2 signals were used without across-gene standardization or rounding.

Estimands and prioritization

For each gene and histology we calculated $A = P(\text{EMC} > \text{comparator}) + 0.5P(\text{tie})$ across all available specimen pairs. $A = 0.5$ denotes neutral pair ordering; $A = 0.7$ corresponds to 70% superiority after assigning half of tied pairs. This is neither a fold change nor a diagnostic classification accuracy established in an external test set.

We reported marginal A using all nine EMC and, separately, exact-sequencing-year contrasts. Years with at least one EMC and one relevant comparator were retained, including singleton strata. Within each histology, supported-year effects were weighted by the number of EMC in those years. MLPS/LGFMS supported 2019 and 2021; synovial sarcoma supported 2019 and 2020. Thus each matched contrast used three EMC, with four unique patients across the three contrasts. The

marginal and matched summaries answer different questions and were not substituted for one another.

A gene met the prospective-to-us tissue-prioritization rule only if its marginal equal-histology A was ≥ 0.70 , each primary histology had $A > 0.50$ under both definitions, and both composite directions remained > 0.50 after every single-EMC and single-histology deletion. The 0.70 benchmark was frozen as a tangible 20-percentage-point departure from neutral ordering, not selected from new values and not a clinical cutoff. Sequencing-year, comparator-patient and revised-diagnosis sensitivities were prespecified interpretation qualifiers, not additional pass criteria. No rule was applied to CHRNA6.

The primary cross-cohort anchor was LGFMS-specific A, calculated separately in each cohort. MFS, SFT and desmoid were separate secondary shared-histology comparisons. We compared effect size, sign and single-biopsy deletion sensitivity, without pooling platform values, defining another success cutoff or substituting a different comparator composite. GSE4303 was excluded from abundance replication because the composition of its two-color reference was unresolved.

Uncertainty and reproducibility

For primary Hofvander estimates, 2000 bootstrap resamples (seed 20260906) were stratified by histology and sequencing year, sharing each EMC resample across comparisons. Reported 2.5th–97.5th percentiles are pointwise intervals conditional on observed strata. Singleton strata remain fixed and can produce overly narrow or degenerate intervals. We therefore also report raw denominators, individual placements and all single-patient, histology and year deletions. Removing a year or last specimen can change support; weights were recalculated and unsupported contrasts remained undefined. No P values or multiplicity-selected discovery set was produced.

HPA gene XML/JSON records supplied separate normal-expression and localization context.[6, 7] Version 25 XML entries and current 25.1 documentation were distinguished. Normal IHC reliability, absent records, compartment discordance and antibody ambiguity were retained. No HPA nTPM-to-tumor-TPM safety ratio was calculated. Historical GSE28866 findings were considered under their distinct depth-scaled, square-root-compressed 3SEQ peak estimand, without pooling values or recounting four EMC library records as four established independent patients.[8]

The sample/probe contracts and prioritization rule were frozen before new target-expression values were inspected. Source files, metadata and code are hash-pinned. An independent implementation reparsed the original arrays, platform annotation and RNA-sequencing matrix and reproduced the estimates and deletion calculations. The supplement gives the exact files and remaining provenance limitations.

Use of artificial intelligence. OpenAI Codex assisted with source recovery and organization, analysis-code implementation, execution, verification, and manuscript drafting. Independent implementations checked original source values and the reported arithmetic. AI assistance does not establish biological validity; responsibility for the final interpretation and released article remains with the author.

Results

Panel results depend on comparator histology

CSPG4 alone met the fixed-panel rule, with marginal composite $A=0.89454$ (pointwise conditional interval 0.86614–0.91806) and matched $A=0.81111$ (0.75278–0.86111). Its marginal A values were 0.78571 versus MLPS, 0.96581 versus LGFMS and 0.93210 versus synovial sarcoma; corresponding matched values were 0.66667, 0.93333 and 0.83333. CHRNA6 had $A=1$ across these comparisons but remained outside the 11-gene pass count. Figure 1 and Table S1 display every gene, including reversed and neutral directions.

PRAME and L1CAM had positive LGFMS directions in both datasets but failed broad prioritization because other Hofvander histologies had lower or reversed contrasts. For PRAME, marginal A was 0 versus MLPS and 0.00617 versus synovial sarcoma. L1CAM's marginal composite was 0.69931, near but below 0.70; importantly, its synovial comparison was independently nonpositive (0.43827 marginal; 0.32500 matched), so its failure did not depend solely on a close numerical threshold.

MSLN, SSTR2, GPC3 and FAP had opposed GSE24369-versus-Hofvander LGFMS directions. CD276, CD248 and CDH17 were below 0.5 against LGFMS in both cohorts. ALPP had positive marginal LGFMS ordering but matched $A=0.5$. These contrasts are retained rather than interpreted through the most favorable histology.

CSPG4 ordering is consistent against LGFMS but sensitive to sequencing year

Against LGFMS, CSPG4 A was 1.000 in the arrays and 0.96581 marginal/0.93333 matched in Hofvander. Every single array EMC or LGFMS biopsy deletion retained $A=1$. Secondary shared contexts also had positive CSPG4 directions: array $A=1$ versus each of MFS, SFT and desmoid; Hofvander marginal/matched values were 0.85370/0.85714, 0.87879/0.62500 and 1/1 respectively. These comparisons support consistent ordering against those particular histologies, not universal specificity.

Within Hofvander, single-EMC deletion composite ranges were 0.88136–0.94035 marginal and 0.71667–0.94444 matched. Single-histology deletion ranges were 0.85891–0.94896 and 0.75000–

0.88333. However, deleting 2019 lowered the matched composite to 0.43333 while the marginal value remained 0.83333 (Figure 2). The apparently positive matched summary therefore depends on a small set of supported year cells; it cannot be called batch-robust. The conditional bootstrap does not resolve that limitation. Revised diagnoses gave similar composites (0.89475 marginal; 0.81111 matched), but partly expression-informed revisions are not independent confirmation.

CSPG4 was not higher than DFSP marginally ($A=0.46667$). The matched DFSP comparison was especially sparse, with one EMC and three DFSP specimens in 2021. This contrary histology was not included in the primary success rule and remains consequential context.

Protein and normal expression evidence limits interpretation

The retrieved HPA CSPG4 record describes broad normal cytoplasmic IHC, whereas ICC/IF includes membrane localization. Different assay compartments cannot establish EMC cell-surface density or normal sparing. Historical GSE28866 CSPG4 peak medians were broadly positive, yet a normal-colon record exceeded the lowest EMC library value. That observation uses a different estimand, but directly argues against treating positive group medians as universal individual separation. Historical CHRNA6 peak signal was nonuniform; it neither invalidates the published RNA CISH assay nor implies absence of the whole transcript.

Independent protein evidence further limits RNA-only extrapolation. Cammareri et al. assessed PRAME clone QR005 on whole sections from 350 soft-tissue tumors and mimics.[9] Their original supplementary rows 128–132 contain five EMC cases, all negative in both reader columns; four have recorded fusion/rearrangement support and one has no recorded ancillary test. Negative meant 0% positive cells. PRAME expression was common in MLPS and synovial sarcoma in that study. Positive PRAME ranks relative to LGFMS therefore do not establish PRAME protein positivity or an EMC therapeutic address. These protein observations come from a separate cohort and assay, not a paired RNA–protein experiment.

Discussion

The main contribution is a comparative molecular characterization of EMC that identifies both a CSPG4 tissue-validation question and the limits of its RNA rationale. CSPG4 has positive ordering against LGFMS in both cohorts and against the specified primary RNA-sequencing comparators. Yet the matched summary rests on four unique EMC patients and reverses when 2019 is removed. Together with the nonpositive DFSP comparison, this makes comparator and year dependence central to the finding. The next biological question is whether CSPG4 protein is present in malignant cells and accessible in EMC tissue across relevant technical and biological variation.

The fixed panel also produces useful negative evidence. Some transcripts separate EMC from LGFMS while failing against MLPS or synovial sarcoma; others change direction across platforms. Preserving these results avoids presenting the preferred comparator as if it represented sarcoma generally. CHRNA6 recapitulates a previously supported RNA marker and is deliberately not counted as a new target. In particular, neither its established CISH performance nor the present rank separation demonstrates protein immunoreactivity or therapeutic accessibility.

Several limitations remain. The source cohort was a convenience sample, and sequencing year is an incomplete proxy for library chemistry, RNA quality and processing batch. Restricting to supported years substantially reduces the EMC sample. The array biopsy lesion stage is unknown, and exclusion of explicit old cases cannot prove absence of every historical overlap. Bulk tissue admixture can produce differences unrelated to malignant-cell expression; EMC purity and grade were unavailable for adjustment. TPM and RMA signals have different measurement properties despite a common rank estimand. Normal tissues were not matched to tumors, HPA includes incomplete or uncertain protein evidence, and the two muscle pools cannot establish organ safety. Sex- and gender-specific effects were not assessed; these small cohorts do not establish generalizability across sex or gender groups. Finally, the prioritization threshold is an allocation rule rather than a clinically validated effect size or a statistical discovery procedure.

The complete panel, including its negative and discordant results, provides a reproducible basis for that targeted tissue question. Demonstrating malignant-cell localization, protein accessibility, diagnostic performance or a therapeutic window would require evidence beyond this RNA reanalysis.

Data code and declarations

Data availability. Original measurements and metadata are available from the Hofvander author release and Zenodo record (<https://doi.org/10.5281/zenodo.17866629>), GEO GSE24369/GPL6244 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24369>), GEO GSE28866 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28866>) and the archived Human Protein Atlas entries identified in the supplement. The complete code/data supplement is publicly deposited with Research Square version 1 [4]: <https://assets-eu.researchsquare.com/files/rs-10959636/v1/b95606f3a3799c60c8ac4434.zip>. It contains the frozen protocols, original source hashes, specimen/probe mappings, analysis code, full-panel effect tables and offline replay instructions. This revision changes presentation and documentation; it retains the same computations and archived evidence. The manuscript license does not replace source-specific attribution or reuse conditions in that archive.

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Research scope and declarations. This work reanalyzes public data and reports no newly collected participant specimens, intervention, protein experiment or clinical treatment. No new ethics approval, exemption or waiver is asserted.

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References

- [1] Dulken BW, Kingsley L, Zdravkovic S, Cespedes O, Qian X, Suster DI, et al. CHRNA6 RNA In Situ Hybridization Is a Useful Tool for the Diagnosis of Extraskeletal Myxoid Chondrosarcoma. *Mod Pathol*. 2024;37(5):100464. <https://doi.org/10.1016/j.modpat.2024.100464>.
- [2] Hofvander J, Köster J, Sydow S, Piccinelli P, Vult von Steyern F, Tsagkosis P, et al. Transcriptomic Subgroups in Soft Tissue Tumors Correlate with Morphologic Subtype, Genomic Features, and Outcome. *Clin Cancer Res*. 2026;32(9):1825-1834. <https://doi.org/10.1158/1078-0432.CCR-25-3740>.
- [3] McRae TD. EMC research program expression-panel definitions. Curated surface_antigen groups and route_named_addresses in emc_expression_panels.py. Pinned repository revision 81d2931733aae640220c042039d0c53efed1690a. https://github.com/trimcrae/Rare-cancers/blob/81d2931733aae640220c042039d0c53efed1690a/research/modalities/emc_expression_panels.py (accessed 9 September 2026).
- [4] McRae TD. Fixed-panel tissue RNA prioritization in extraskeletal myxoid chondrosarcoma: CSPG4 evidence across cohorts with comparator and sequencing-year limits. Research Square. 2026. Version 1 and accompanying code/data supplement [preprint and dataset]. <https://doi.org/10.21203/rs.3.rs-10959636/v1>.
- [5] National Center for Biotechnology Information. Gene Expression Omnibus series GSE24369 and platform GPL6244. Original series data and transcript-cluster annotation. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24369>; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GPL6244> (accessed 9 September 2026).
- [6] Uhlén M, Fagerberg L, Hallström BM, Lindskog C, Oksvold P, Mardinoglu A, et al. Tissue-based map of the human proteome. *Science*. 2015;347(6220):1260419. <https://doi.org/10.1126/science.1260419>.
- [7] Thul PJ, Åkesson L, Wiking M, Mahdessian D, Geladaki A, Ait Blal H, et al. A subcellular map of the human proteome. *Science*. 2017;356(6340):eaal3321. <https://doi.org/10.1126/science.aal3321>.

[8] Brunner AL, Beck AH, Edris B, Sweeney RT, Zhu SX, Li R, et al. Transcriptional profiling of long non-coding RNAs and novel transcribed regions across a diverse panel of archived human cancers. *Genome Biol.* 2012;13(8):R75. <https://doi.org/10.1186/gb-2012-13-8-r75>. Data: GEO GSE28866.

[9] Cammareri C, Beltzung F, Michal M, Vanhersecke L, Coindre JM, Velasco V, et al. PRAME immunohistochemistry in soft tissue tumors and mimics: a study of 350 cases highlighting its imperfect specificity but potentially useful diagnostic applications. *Virchows Arch.* 2023;483(2):145-156. <https://doi.org/10.1007/s00428-023-03606-6>.

Figure legends

Figure 1. Fixed-panel probability of superiority by comparator and cohort. Rows retain all 11 address genes plus the separately marked CHRNA6 control. Panel A shows the same LGFMS comparison in original arrays (6 EMC, 17 LGFMS), Hofvander marginal (9 EMC, 13 LGFMS), and Hofvander year-matched (3 EMC, 12 LGFMS across 2019/2021). Panels B and C show all three primary Hofvander histologies separately, marginally and within supported years. Color and printed values encode A on 0–1 with a neutral midpoint 0.5; there are no significance stars. The matched columns do not represent nine independent matched patients.

Figure 2. CSPG4 specimens and sequencing-year sensitivity. Panel A shows each retained primary specimen's $\log_2(1+TPM)$, grouped by EMC and the three primary comparator histologies; color denotes sequencing year. The display transformation does not enter rank calculations. Panel B shows the matched equal-histology summary with each year deleted, beside the full-data value and neutral 0.5 reference. The 2019 deletion reverses the summary. Points are descriptive observed/deletion values, not confidence limits; no normal-tissue or protein measurements are plotted.

Figure 1

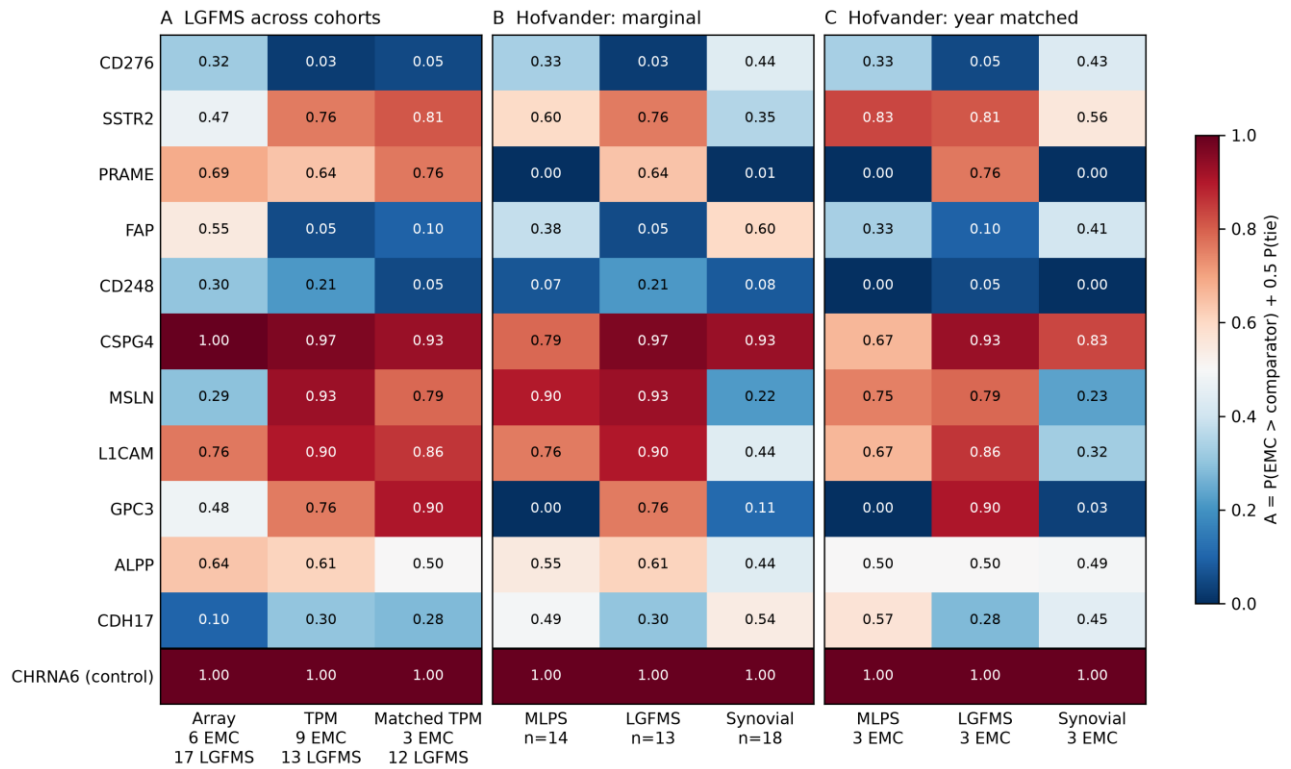


Figure 2

